

Loop Quantum Gravity

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1 A Review of Einstein-Hilbert Action

Consider the action

$$S_{\text{EH}} = \frac{c^4}{8\pi G} \int d^4x \sqrt{-g} R + S_{\text{matter}}$$

Using functional derivative with respect to $g_{\mu\nu}$, we can derive the Einstein equation.

$$\begin{aligned} \delta S_{\text{EH}} &= \frac{c^4}{8\pi G} \int d^4x \delta (\sqrt{-g} g^{\mu\nu} R_{\mu\nu}) + \delta S_{\text{matter}} \\ &= \frac{c^4}{8\pi G} \int d^4x [(\delta \sqrt{-g}) R + \sqrt{-g} (\delta g^{\mu\nu}) R_{\mu\nu} + \sqrt{-g} g^{\mu\nu} (\delta R_{\mu\nu})] + \delta S_{\text{matter}} \end{aligned}$$

The first term uses the Jacobi's formula.

$$\delta g = g g^{\mu\nu} \delta g_{\mu\nu}$$

We can prove it by an invertible matrix A .

$$\det A = \sum_j A_{ij} (\text{adj } A)_{ji}$$

Noting that $(\text{adj } A)_{ji}$ has no dependence on A_{ik} and A_{kj} for any k , we see

$$\frac{\partial}{\partial A_{ik}} \det A = \sum_j \delta_{jk} (\text{adj } A)_{ji} + \sum_j A_{ij} \left[\frac{\partial}{\partial A_{ik}} (\text{adj } A)_{ji} \right] = (\text{adj } A)_{ki}$$

$$d(\det A) = \sum_{i,j} \frac{\partial \det A}{\partial A_{ij}} dA_{ij} = \sum_{i,j} (\text{adj } A)_{ji} dA_{ij}$$

For the metric $g_{\mu\nu} = A_{ij}$, we have $g = \det A$ and $g^{\mu\nu} = \frac{(\text{adj } A)_{ji}}{\det A}$, so the formula becomes

$$\delta g = g g^{\mu\nu} \delta g_{\mu\nu}$$

Then, we have

$$\delta \sqrt{-g} = -\frac{\delta g}{2\sqrt{-g}} = \frac{1}{2} \sqrt{-g} g^{\mu\nu} \delta g_{\mu\nu} = -\frac{1}{2} \sqrt{-g} g_{\mu\nu} \delta g^{\mu\nu}$$

The third term in δS_{EH} vanishes. To see that, we should know that $\delta\Gamma_{\mu\nu}^\alpha$ is a tensor.

$$\begin{aligned}\delta R^\rho_{\sigma\mu\nu} &= \partial_\mu \delta\Gamma^\rho_{\nu\sigma} - \partial_\nu \delta\Gamma^\rho_{\mu\sigma} + \delta\Gamma^\alpha_{\nu\sigma} \Gamma^\rho_{\mu\alpha} + \Gamma^\alpha_{\nu\sigma} \delta\Gamma^\rho_{\mu\alpha} - \delta\Gamma^\alpha_{\mu\sigma} \Gamma^\rho_{\nu\alpha} - \Gamma^\alpha_{\mu\sigma} \delta\Gamma^\rho_{\nu\alpha} \\ &= (\partial_\mu \delta\Gamma^\rho_{\nu\sigma} + \Gamma^\rho_{\mu\alpha} \delta\Gamma^\alpha_{\nu\sigma} - \Gamma^\alpha_{\mu\nu} \delta\Gamma^\rho_{\alpha\sigma} - \Gamma^\alpha_{\mu\sigma} \delta\Gamma^\rho_{\nu\alpha}) - (\partial_\nu \delta\Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\nu\alpha} \delta\Gamma^\alpha_{\mu\sigma} - \Gamma^\alpha_{\nu\sigma} \delta\Gamma^\rho_{\mu\alpha} - \Gamma^\alpha_{\nu\mu} \delta\Gamma^\rho_{\alpha\sigma}) \\ &= \nabla_\mu (\delta\Gamma^\rho_{\nu\sigma}) - \nabla_\nu (\delta\Gamma^\rho_{\mu\sigma})\end{aligned}$$

Hence, we have

$$\delta R_{\sigma\nu} = \delta R^\mu_{\sigma\mu\nu} = \nabla_\mu (\delta\Gamma^\mu_{\nu\sigma}) - \nabla_\nu (\delta\Gamma^\mu_{\mu\sigma})$$

Using the compatibility of the metric $\partial_\alpha g^{\mu\nu} = \partial_\alpha \sqrt{-g} = 0$ and the divergence theorem, the integral for the third term vanishes as $\delta g_{\mu\nu} = 0$ at the boundary.

$$\begin{aligned}\int d^4x \sqrt{-g} g^{\mu\nu} \delta R_{\mu\nu} &= \int d^4x [\nabla_\sigma (\sqrt{-g} g^{\mu\nu} \delta\Gamma^\sigma_{\mu\nu}) - \nabla_\nu (\sqrt{-g} g^{\mu\nu} \delta\Gamma^\sigma_{\sigma\mu})] \\ &= \int d^4x [\partial_\sigma (\sqrt{-g} g^{\mu\nu} \delta\Gamma^\sigma_{\mu\nu}) - \partial_\nu (\sqrt{-g} g^{\mu\nu} \delta\Gamma^\sigma_{\sigma\mu})] \\ &= 0\end{aligned}$$

For the matter term

$$S_{\text{matter}} = \int d^4x \sqrt{-g} \mathcal{L}_M \quad \Rightarrow \quad \delta S_{\text{matter}} = \int d^4x \delta (\sqrt{-g} \mathcal{L}_M)$$

Defining the energy-stress tensor

$$T_{\mu\nu} = -\frac{1}{\sqrt{-g}} \frac{\delta (\sqrt{-g} \mathcal{L}_M)}{\delta g^{\mu\nu}}$$

The functional derivative now becomes

$$\delta S_{EH} = \frac{c^4}{8\pi G} \int d^4x \sqrt{-g} \delta g^{\mu\nu} \left[-\frac{1}{2} g_{\mu\nu} R + R_{\mu\nu} - \frac{8\pi G}{c^4} T_{\mu\nu} \right]$$

For $\delta S_{EH} = 0$, we see that Einstein equation always holds

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = \frac{8\pi G}{c^4} T_{\mu\nu}$$

2 Arnowitt–Deser–Misner (ADM) Formalism

To foliate the spacetime into a family of spacelike hypersurfaces

$$\Sigma_{t_0} = \{x^\mu \mid t(x^\mu) = t_0\}$$

labeled by their time coordinate t_0 , so the covector $\partial_\mu t$ is normal to the hypersurface Σ_t . We define the basis vectors $t^\mu = \left(\frac{\partial}{\partial t}\right)^\mu$ and $e_i^\mu = \left(\frac{\partial}{\partial x_i}\right)^\mu$ by

$$t^\mu \partial_\mu t = 1 \quad \text{and} \quad e_i^\mu \partial_\mu t = 0$$

In general, the vector $t^\mu = e_t^\mu$ is not normal to the hypersurface Σ_t . We can define the unit normal covector

$$n_\mu = -N\partial_\mu t \quad \text{and} \quad n_\mu n^\mu = -1, \quad e_i^\mu n_\mu = -N(e_i^\mu \partial_\mu t) = 0$$

where N is called the lapse function. The negative sign is chosen, so n^μ is future-directed. Then, the vector t^μ can be decompose into

$$t^\mu = Nn^\mu + N^i e_i^\mu$$

where $N^i e_i^\mu$ is the shift vector tangent to the hypersurface. We can then compute the metric

$$g_{00} = g_{\mu\nu} e_t^\mu e_t^\nu = g_{\mu\nu} (Nn^\mu + N^\mu) (Nn^\nu + N^\nu) = -N^2 + q_{ij} N^i N^j$$

$$g_{0i} = g_{\mu\nu} e_t^\mu e_i^\nu = g_{\mu\nu} (Nn^\mu + N^\mu) e_i^\nu = q_{ij} N^j$$

$$g_{ij} = g_{\mu\nu} e_i^\mu e_j^\nu = q_{ij}$$

where q_{ij} is just the space metric to the 3-geometry of the hypersurface Σ_t . The line segment is

$$\begin{aligned} ds^2 &= (-N^2 + q_{ij} N^i N^j) dt^2 + 2q_{ij} N^j dt dx^i + q_{ij} dx^i dx^j \\ &= -N^2 dt^2 + q_{ij} (N^i dt + dx^i) (N^j dt + dx^j) \end{aligned}$$

We can also derive the inverse metric $g^{\mu\nu}$ by solving $g_{\mu\sigma} g^{\sigma\nu} = \delta_\mu^\nu$

$$\begin{aligned} 0 &= \delta_i^0 = g_{i0} g^{00} + g_{ij} g^{j0} \\ &= q_{ij} N^j g^{00} + q_{ij} g^{j0} \\ &= q_{ij} (N^j g^{00} + g^{j0}) \Rightarrow g^{j0} = -N^j g^{00} \end{aligned}$$

$$\begin{aligned} 1 &= \delta_0^0 = g_{00} g^{00} + g_{0i} g^{i0} \\ &= (-N^2 + q_{ij} N^i N^j) g^{00} - q_{ij} N^i N^j g^{00} \\ &= -N^2 g^{00} \Rightarrow g^{00} = -\frac{1}{N^2}, \quad g^{0i} = \frac{N^i}{N^2} \end{aligned}$$

$$\begin{aligned} 0 &= \delta_0^k = g_{00} g^{0k} + g_{0j} g^{jk} \\ &= (-N^2 + q_{ij} N^i N^j) \frac{N^k}{N^2} + q_{ij} N^i g^{jk} \\ &= q_{ij} N^i \left(-q^{jk} + \frac{N^j N^k}{N^2} - g^{jk} \right) \Rightarrow g^{jk} = -q^{jk} + \frac{N^j N^k}{N^2} \end{aligned}$$

Before the end of the discussion, we define the projected metric

$$h_{\mu\nu} = q_{ij} e_\mu^i e_\nu^j$$

For any vector V^μ expressed by

$$V^\mu = V^i e_i^\mu + V^n n^\mu$$

The projection can be calculated by

$$\begin{aligned} V^\mu e_\mu^i &= (V^j e_j^\mu + V^n n^\mu) e_\mu^i = V^j e_j^\mu (g_{\mu\nu} q^{ik} e_k^\nu) = V^j q^{ik} q_{jk} = V^j \delta_j^i = V^i \\ -V^\mu n_\mu &= -(V^j e_j^\mu + V^n n^\mu) n_\mu = -V^n n^\mu n_\mu = V^n \end{aligned}$$

Hence, if we write

$$V^\nu \delta^\mu_\nu = V^\mu = (V^\nu e_\nu^i) e_i^\mu + (-V^\nu n_\nu) n^\mu = V^\nu (e_i^\mu e_\nu^i - n^\mu n_\nu) = V^\nu (h^\mu_\nu - n^\mu n_\nu)$$

The spatial projection metric can be expressed by

$$h_{\mu\nu} = g_{\mu\nu} + n_\mu n_\nu$$

3 Extrinsic Curvature

For a vector $V^\mu = V^i e_i^\mu$ tangent to the hypersurface Σ_t , the covariant derivative to the 3-geometry is

$${}^{(3)}\nabla_i V^j = \partial_i V^j + {}^{(3)}\Gamma_{ik}^j V^k$$

where ${}^{(3)}\Gamma_{ij}^k$ is the connecting coefficients on the hypersurface Σ_t embedded in a 4D manifold $\mathcal{M}$. The result stays tangent to the hypersurface Σ_t . In the language of vector, we express this by projections.

$$e_\mu^k {}^{(3)}\nabla_i V^\mu = \partial_i V^k + {}^{(3)}\Gamma_{ij}^k V^j$$

In a special case $V^\mu = e_j^\mu$, we obtain an expression for ${}^{(3)}\Gamma_{ij}^k$

$$e_\mu^k {}^{(3)}\nabla_i e_j^\mu = {}^{(3)}\Gamma_{ij}^k$$

However, the covariant derivative to the 4-geometry is not tangent to the hypersurface anymore. Taking the covariant derivative of the normal condition,

$$n_\nu V^\nu = 0 \quad \Rightarrow \quad 0 = {}^{(4)}\nabla_i (n_\nu V^\nu) = ({}^{(4)}\nabla_i n_\nu) V^\nu + n_\nu ({}^{(4)}\nabla_i V^\nu)$$

The normal component of ${}^{(4)}\nabla_i V^\nu$ can then be expressed by

$$n_\nu ({}^{(4)}\nabla_i V^\nu) = - ({}^{(4)}\nabla_i n_\nu) V^\nu = - (e_i^\mu {}^{(4)}\nabla_\mu n_\nu) (V^j e_j^\nu) = (-e_i^\mu e_j^\nu {}^{(4)}\nabla_\mu n_\nu) V^j = K_{ij} V^j$$

where we defined the extrinsic curvature

$$K_{ij} = -e_i^\mu e_j^\nu {}^{(4)}\nabla_\mu n_\nu = -e_j^\nu {}^{(4)}\nabla_i n_\nu$$

Additionally, since $n_\nu e_j^\nu = 0$, we can also write

$$e_j^\nu {}^{(4)}\nabla_i n_\nu = -n_\nu {}^{(4)}\nabla_i e_j^\nu \quad \Rightarrow \quad K_{ij} = n_\nu {}^{(4)}\nabla_i e_j^\nu$$

The extrinsic curvature K_{ij} is exactly the normal projection of ${}^{(4)}\nabla_i e_j^\nu$. On the other hand, the intrinsic covariant derivative ${}^{(3)}\nabla_i e_j^\mu$ is the tangential projection of the spacetime covariant derivative ${}^{(4)}\nabla_i e_j^\mu$ by definition. Hence,

$$e_\mu^k {}^{(4)}\nabla_i e_j^\mu = e_\mu^k {}^{(3)}\nabla_i e_j^\mu = {}^{(3)}\Gamma_{ij}^k$$

Combining with the normal component, we obtain the Gauss decomposition formula.

$${}^{(4)}\nabla_i e_j^\nu = {}^{(3)}\Gamma_{ij}^k e_k^\nu - K_{ij} n^\nu$$

Therefore, the covariant derivative of the vector V^μ can be decomposed by

$$\begin{aligned} {}^{(4)}\nabla_i V^\nu &= {}^{(4)}\nabla_i (V^j e_j^\nu) \\ &= e_j^\nu \partial_i V^j + V^j {}^{(4)}\nabla_i e_j^\nu \\ &= e_j^\nu \partial_i V^j + V^j ({}^{(3)}\Gamma_{ij}^k e_k^\nu - K_{ij} n^\nu) \\ &= e_j^\nu (\partial_i V^j + {}^{(3)}\Gamma_{ik}^j V^k) - K_{ij} V^j n^\nu \\ &= e_j^\nu {}^{(3)}\nabla_i V^j - K_{ij} V^j n^\nu \end{aligned}$$

In the adapted coordinates $n_\mu = (-N, 0, 0, 0)$, the extrinsic curvature is

$$\begin{aligned} K_{ij} &= -e_i^\mu e_j^\nu {}^{(4)}\nabla_\mu n_\nu \\ &= -e_i^\mu e_j^\nu \partial_\mu n_\nu + e_i^\mu e_j^\nu {}^{(4)}\Gamma_{\mu\nu}^\alpha n_\alpha \\ &= -N {}^{(4)}\Gamma_{ij}^0 \\ &= -\frac{N}{2} [g^{00} (\partial_i g_{0j} + \partial_j g_{0i} - \partial_0 g_{ij}) + g^{0k} (\partial_i g_{jk} + \partial_j g_{ik} - \partial_k g_{ij})] \\ &= -\frac{N}{2} \left[-\frac{1}{N^2} (\partial_i N_j + \partial_j N_i - \dot{q}_{ij}) + \frac{N^k}{N^2} (\partial_i q_{jk} + \partial_j q_{ik} - \partial_k q_{ij}) \right] \\ &= \frac{1}{2N} [(\partial_i N_j + \partial_j N_i - \dot{q}_{ij}) - N_k {}^{(3)}\Gamma_{ij}^k] \\ &= \frac{1}{2N} ({}^{(3)}\nabla_i N_j + {}^{(3)}\nabla_j N_i - \dot{q}_{ij}) \end{aligned}$$

Moreover, using $n^{\nu(4)}\nabla_\mu n_\nu = 0$ and $K_{ij} = K_{ji}$, the scalars constructed by K_{ij} can be expressed by

$$K = q^{ij} K_{ij} = -q^{ij} e_i^\mu e_j^\nu {}^{(4)}\nabla_\mu n_\nu = -h^{\mu\nu} {}^{(4)}\nabla_\mu n_\nu = -(g^{\mu\nu} + n^\mu n^\nu) {}^{(4)}\nabla_\mu n_\nu = -{}^{(4)}\nabla_\mu n^\mu$$

$$\begin{aligned} K_{ij} K^{ij} &= K_{ij} K^{ji} \\ &= e_i^\mu e_j^\nu e_\alpha^i e_\beta^j {}^{(4)}\nabla_\mu n_\nu {}^{(4)}\nabla^\beta n^\alpha \\ &= h^\mu{}_\alpha h^\nu{}_\beta {}^{(4)}\nabla_\mu n_\nu {}^{(4)}\nabla^\beta n^\alpha \\ &= (\delta^\mu{}_\alpha + n^\mu n_\alpha) (\delta^\nu{}_\beta + n^\nu n_\beta) {}^{(4)}\nabla_\mu n_\nu {}^{(4)}\nabla^\beta n^\alpha \\ &= {}^{(4)}\nabla_\mu n_\nu {}^{(4)}\nabla^\nu n^\mu \end{aligned}$$

4 Curvature Tensor and Ricci Scalar

Recalling that we can calculate the curvature tensor by

$$[\nabla_\alpha, \nabla_\beta]V^\mu = R^\mu{}_{\nu\alpha\beta}V^\nu$$

we first compute

$$\begin{aligned} {}^{(4)}\nabla_i {}^{(4)}\nabla_j e_k^\mu &= {}^{(4)}\nabla_i \left({}^{(3)}\Gamma_{jk}^a e_a^\mu - K_{jk} n^\mu \right) \\ &= \left[(\partial_i {}^{(3)}\Gamma_{jk}^a) e_a^\mu + {}^{(3)}\Gamma_{jk}^b {}^{(3)}\Gamma_{ib}^a e_a^\mu - K_{ia} {}^{(3)}\Gamma_{jk}^a n^\mu \right] - [(\partial_i K_{jk}) n^\mu - K_{jk} K_i^a e_a^\mu] \\ &= (\partial_i {}^{(3)}\Gamma_{jk}^a + {}^{(3)}\Gamma_{jk}^b {}^{(3)}\Gamma_{ib}^a + K_{jk} K_i^a) e_a^\mu - (K_{ia} {}^{(3)}\Gamma_{jk}^a + \partial_i K_{jk}) n^\mu \end{aligned}$$

For $[{}^{(4)}\nabla_i, {}^{(4)}\nabla_j] e_k^\mu$, we express the curvature to the 3-geometry by

$${}^{(3)}R^a{}_{kij} = \partial_i {}^{(3)}\Gamma_{jk}^a - \partial_j {}^{(3)}\Gamma_{ik}^a + {}^{(3)}\Gamma_{jk}^b {}^{(3)}\Gamma_{ib}^a - {}^{(3)}\Gamma_{ik}^b {}^{(3)}\Gamma_{jb}^a$$

we can write

$$\begin{aligned} [{}^{(4)}\nabla_i, {}^{(4)}\nabla_j] e_k^\mu &= ({}^{(3)}R^a{}_{kij} + K_{jk} K_i^a - K_{ik} K_j^a) e_a^\mu - (K_{ia} {}^{(3)}\Gamma_{jk}^a - K_{ja} {}^{(3)}\Gamma_{ik}^a + \partial_i K_{jk} - \partial_j K_{ik}) n^\mu \\ &= ({}^{(3)}R^a{}_{kij} + K_{jk} K_i^a - K_{ik} K_j^a) e_a^\mu - ({}^{(3)}\nabla_i K_{jk} - {}^{(3)}\nabla_j K_{ik}) n^\mu \end{aligned}$$

Since $[{}^{(4)}\nabla_i, {}^{(4)}\nabla_j] e_k^\mu = {}^{(4)}R^\mu{}_{\nu ij} e_k^\nu = {}^{(4)}R^\mu{}_{kij}$, we project the tensor by

$$\begin{aligned} {}^{(4)}R^l{}_{kij} &= {}^{(4)}R^\mu{}_{kij} e_\mu^l = {}^{(3)}R^l{}_{kij} + K_{jk} K_i^l - K_{ik} K_j^l \\ {}^{(4)}R^n{}_{kij} &= -{}^{(4)}R^\mu{}_{kij} n_\mu = {}^{(3)}\nabla_j K_{ik} - {}^{(3)}\nabla_i K_{jk} \end{aligned}$$

We can contract the indices, which gives

$${}^{(4)}R^{ij}{}_{ij} = {}^{(3)}R^{ij}{}_{ij} + K^2 - K_{ij} K^{ij}$$

With the same trick, we can calculate

$$\begin{aligned} R^{in}{}_{in} &= -e_\mu^i e_i^\alpha n^\beta (R^\mu{}_{\nu\alpha\beta} n^\nu) \\ &= -e_\mu^i e_i^\alpha n^\beta [{}^{(4)}\nabla_\alpha, {}^{(4)}\nabla_\beta] n^\mu \\ &= -(\delta_\mu^\alpha + n_\mu n^\alpha) n^\beta [{}^{(4)}\nabla_\alpha, {}^{(4)}\nabla_\beta] n^\mu \\ &= -n^\beta [{}^{(4)}\nabla_\alpha, {}^{(4)}\nabla_\beta] n^\alpha \\ &= -n^\beta ({}^{(4)}\nabla_\alpha {}^{(4)}\nabla_\beta n^\alpha) + n^\beta ({}^{(4)}\nabla_\beta {}^{(4)}\nabla_\alpha n^\alpha) \\ &= -{}^{(4)}\nabla_\alpha (n^\beta {}^{(4)}\nabla_\beta n^\alpha) + {}^{(4)}\nabla_\beta (n^\beta {}^{(4)}\nabla_\alpha n^\alpha) + K_{ij} K^{ij} - K^2 \\ &= -{}^{(4)}\nabla_\alpha (n^\beta {}^{(4)}\nabla_\beta n^\alpha - n^\alpha {}^{(4)}\nabla_\beta n^\beta) + K_{ij} K^{ij} - K^2 \\ R^{nn}{}_{nn} &= -n_\mu n^\alpha n^\beta (R^\mu{}_{\nu\alpha\beta} n^\nu) = -n_\mu n^\alpha n^\beta [{}^{(4)}\nabla_\alpha, {}^{(4)}\nabla_\beta] n^\mu = 0 \end{aligned}$$

The Ricci scalar is then given by

$$\begin{aligned} {}^{(4)}R &= {}^{(4)}R^{\mu\nu}{}_{\mu\nu} \\ &= {}^{(4)}R^{ij}{}_{ij} + 2{}^{(4)}R^{in}{}_{in} + R^{nn}{}_{nn} \\ &= {}^{(3)}R^{ij}{}_{ij} - K^2 + K_{ij} K^{ij} - 2{}^{(4)}\nabla_\alpha (n^\beta {}^{(4)}\nabla_\beta n^\alpha - n^\alpha {}^{(4)}\nabla_\beta n^\beta) \end{aligned}$$

5 Hamiltonian Formulation

We can now express the Einstein-Hilbert action in these coordinates. First, the Jacobian of the metric is

$$\sqrt{-g} = N\sqrt{q}$$

To see this, we can consider the triangular decomposition of a block matrix

$$\begin{aligned} M &= \begin{bmatrix} A & B \\ C & D \end{bmatrix} \\ &= \begin{bmatrix} I & BD^{-1} \\ 0 & I \end{bmatrix} \begin{bmatrix} A - BD^{-1}C & 0 \\ C & D \end{bmatrix} \\ &= \begin{bmatrix} I & BD^{-1} \\ 0 & I \end{bmatrix} \begin{bmatrix} A - BD^{-1}C & 0 \\ 0 & D \end{bmatrix} \begin{bmatrix} I & 0 \\ D^{-1}C & I \end{bmatrix} \end{aligned}$$

Since the determinants of the first and the third matrix are unity, we can then write

$$\det M = \det(A - BD^{-1}C) \det D$$

Taking $M = g_{\mu\nu}$, so that $A = g_{00} = -N^2 + N_i N^i$, $B_i = C_i = N_i$ and $D_{ij} = q_{ij}$, we see

$$g = \det(-N^2 + N_i N^i - q^{ij} N_i N_j) \det(q_{ij}) = -N^2 q \quad \Rightarrow \quad \sqrt{-g} = N\sqrt{q}$$

Neglecting the boundary terms in the curvature ${}^{(4)}R$, we can now write the action

$$S_{EH} = \frac{1}{\kappa} \int d^4x \sqrt{-g} {}^{(4)}R = \frac{1}{\kappa} \int d^4x N\sqrt{q} ({}^{(3)}R - K^2 + K_{ij}K^{ij})$$

So the Lagrangian is

$$\mathcal{L} = \frac{N\sqrt{q}}{\kappa} ({}^{(3)}R - K^2 + K_{ij}K^{ij})$$

Now, taking q_{ij} , N , and N^i as the position variables, the momentum variables are

$$\begin{aligned} \pi^{ij} &= \frac{\partial \mathcal{L}}{\partial \dot{q}_{ij}} = \frac{N\sqrt{q}}{\kappa} \left(\frac{K}{N} q^{ij} - \frac{K^{ij}}{N} \right) = \frac{\sqrt{q}}{\kappa} (K q^{ij} - K^{ij}) \\ p_N &= \frac{\partial \mathcal{L}}{\partial \dot{N}} = 0 \\ p_{\dot{N}^i} &= \frac{\partial \mathcal{L}}{\partial \dot{N}^i} = 0 \end{aligned}$$

By writing

$$\pi = g_{ij} \pi^{ij} = \frac{\sqrt{q}}{\kappa} (3K - K) = \frac{2\sqrt{q}K}{\kappa} \quad \Rightarrow \quad K^{ij} = -\frac{\kappa}{\sqrt{q}} \left(\pi^{ij} - \frac{1}{2} q^{ij} \pi \right)$$

We can then solve for $\dot{q}_{ij}$.

$$\begin{aligned}\dot{q}_{ij} &= {}^{(3)}\nabla_i N_j + {}^{(3)}\nabla_j N_i - 2N K_{ij} \\ &= {}^{(3)}\nabla_i N_j + {}^{(3)}\nabla_j N_i + \frac{2N\kappa}{\sqrt{q}} \left(\pi_{ij} - \frac{1}{2} q_{ij} \pi \right)\end{aligned}$$

The Hamiltonian density is

$$\begin{aligned}\mathcal{H} &= \pi^{ij} \dot{q}_{ij} - \mathcal{L} \\ &= \pi^{ij} \left[{}^{(3)}\nabla_i N_j + {}^{(3)}\nabla_j N_i + \frac{2N\kappa}{\sqrt{q}} \left(\pi_{ij} - \frac{1}{2} q_{ij} \pi \right) \right] - \frac{N\sqrt{q}}{\kappa} ({}^{(3)}R - K^2 + K_{ij} K^{ij}) \\ &= 2\pi^{ij} {}^{(3)}\nabla_i N_j + \frac{2N\kappa}{\sqrt{q}} \left(\pi^{ij} \pi_{ij} - \frac{1}{2} \pi^2 \right) - \frac{N\sqrt{q}}{\kappa} {}^{(3)}R + \frac{N\kappa}{4\sqrt{q}} \pi^2 - \frac{N\kappa}{\sqrt{q}} \left(\pi_{ij} \pi^{ij} - \frac{1}{4} \pi^2 \right) \\ &= 2\pi^{ij} {}^{(3)}\nabla_i N_j + \frac{N\kappa}{\sqrt{q}} (\pi^{ij} \pi_{ij} - \pi^2) - \frac{N\sqrt{q}}{\kappa} {}^{(3)}R\end{aligned}$$

With integral by parts and neglecting the boundary terms, the total Hamiltonian is

$$\begin{aligned}H &= \int_{\Sigma_t} d^3x \mathcal{H} \\ &= \int_{\Sigma_t} d^3x \left(\frac{N\kappa}{\sqrt{q}} (\pi^{ij} \pi_{ij} - \pi^2) - \frac{N\sqrt{q}}{\kappa} {}^{(3)}R + 2\pi^{ij} {}^{(3)}\nabla_i N_j \right) \\ &= \int_{\Sigma_t} d^3x \left[N \left(\frac{\kappa}{\sqrt{q}} (\pi^{ij} \pi_{ij} - \pi^2) - \frac{\sqrt{q}}{\kappa} {}^{(3)}R \right) - 2N_j {}^{(3)}\nabla_i \pi^{ij} \right] \\ &= \int_{\Sigma_t} d^3x (N\mathcal{H}_0 + N_j \mathcal{H}^j)\end{aligned}$$

where

$$\begin{aligned}\mathcal{H}_0 &= \frac{\kappa}{\sqrt{q}} (\pi^{ij} \pi_{ij} - \pi^2) - \frac{\sqrt{q}}{\kappa} {}^{(3)}R \\ \mathcal{H}^j &= -2 {}^{(3)}\nabla_i \pi^{ij}\end{aligned}$$